

A Practical Guide to Uncertainty quantification

TOPIC -- UNCERTAINTY QUANTIFICATION

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d is the dimension of input variables, and r is the dimension of unknown parameters. While $\mathbf{h}_m(\cdot)$ is pre-defined, $\{\beta_m, \sigma_m, \omega_k, k = 1, \dots, d+r\}$, known as hyperparameters of the GP model, need to be estimated via maximum likelihood estimation (MLE). This module can be considered as a generalized kriging method.

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$y_m(\mathbf{x}, \theta) \sim \mathcal{GP}(\mathbf{h}_m(\cdot)^T \beta_m, \sigma_m^2 \mathbf{R}_m(\cdot, \cdot))$

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Aleatoric uncertainty Aleatoric uncertainty is also known as stochastic uncertainty, and is representative of unknowns that differ each time we run the same experiment. For example, a single arrow shot with a mechanical bow that exactly duplicates each launch (the same acceleration, altitude, direction and final velocity) will not all impact the same point on the target due to random and complicated vibrations of the arrow shaft, the knowledge of which cannot be determined sufficiently to eliminate the resulting scatter of impact points. The argument here is in the definition of "cannot": just because we cannot measure sufficiently with our currently available measurement devices does not preclude necessarily the existence of such information, which would move this uncertainty into the category below. Aleatoric is derived from the Latin alea or dice, referring to a game of chance. **Epistemic uncertainty** Epistemic uncertainty is also known as systematic uncertainty, and is due to things one could in principle know but does not in practice. This may be because a measurement is not accurate, because the model neglects certain effects, or because particular data have been deliberately hidden. An example of a source of this uncertainty would be the drag in an experiment designed to measure the acceleration of gravity near the earth's surface. The commonly used gravitational acceleration of 9.8 m/s^2 ignores the effects of air resistance, but the air resistance for the object could be

measured and incorporated into the experiment to reduce the resulting uncertainty in the calculation of the gravitational acceleration. Combined occurrence and interaction of aleatoric and epistemic uncertainty Aleatoric and epistemic uncertainty can also occur simultaneously in a single term E.g., when experimental parameters show aleatoric uncertainty, and those experimental parameters are input to a computer simulation. If then for the uncertainty quantification a surrogate model, e.g. a Gaussian process or a Polynomial Chaos Expansion, is learnt from computer experiments, this surrogate exhibits epistemic uncertainty that depends on or interacts with the aleatoric uncertainty of the experimental parameters. Such an uncertainty cannot solely be classified as aleatoric or epistemic any more, but is a more general inferential uncertainty. In real life applications, both kinds of uncertainties are present. Uncertainty quantification intends to explicitly express both types of uncertainty separately. The quantification for the aleatoric uncertainties can be relatively straightforward, where traditional (frequentist) probability is the most basic form. Techniques such as the Monte Carlo method are frequently used. A probability distribution can be represented by its moments (in the Gaussian case, the mean and covariance suffice, although, in general, even knowledge of all moments to arbitrarily high order still does not specify the distribution function uniquely), or more recently, by techniques such as Karhunen–Loève and polynomial chaos expansions. To evaluate epistemic uncertainties, the efforts are made to understand the (lack of) knowledge of the system, process or mechanism. Epistemic uncertainty is generally understood through the lens of Bayesian probability, where probabilities are interpreted as indicating how certain a rational person could be regarding a specific claim.

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where $y_m(\mathbf{x}, \boldsymbol{\theta})$ denotes the computer model response that depends on several unknown model parameters $\boldsymbol{\theta}$, and $\boldsymbol{\theta}^*$ denotes the true values of the unknown parameters in the course of experiments. The objective is to either estimate $\boldsymbol{\theta}^*$, or to come up with a probability distribution of $\boldsymbol{\theta}^*$ that encompasses the best knowledge of the true parameter values.

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Parameter This comes from the model parameters that are inputs to the computer model (mathematical model) but whose exact values are unknown to experimentalists and cannot be controlled in physical experiments, or whose

values cannot be exactly inferred by statistical methods. Some examples of this are the local free-fall acceleration in a falling object experiment, various material properties in a finite element analysis for engineering, and multiplier uncertainty in the context of macroeconomic policy optimization.

Parametric This comes from the variability of input variables of the model. For example, the dimensions of a work piece in a process of manufacture may not be exactly as designed and instructed, which would cause variability in its performance.

Structural uncertainty Also known as model inadequacy, model bias, or model discrepancy, this comes from the lack of knowledge of the underlying physics in the problem. It depends on how accurately a mathematical model describes the true system for a real-life situation, considering the fact that models are almost always only approximations to reality. One example is when modeling the process of a falling object using the free-fall model; the model itself is inaccurate since there always exists air friction. In this case, even if there is no unknown parameter in the model, a discrepancy is still expected between the model and true physics.

Algorithmic Also known as numerical uncertainty, or discrete uncertainty. This type comes from numerical errors and numerical approximations per implementation of the computer model. Most models are too complicated to solve exactly. For example, the finite element method or finite difference method may be used to approximate the solution of a partial differential equation (which introduces numerical errors). Other examples are numerical integration and infinite sum truncation that are necessary approximations in numerical implementation.

Experimental Also known as observation error, this comes from the variability of experimental measurements. The experimental uncertainty is inevitable and can be noticed by repeating a measurement for many times using exactly the same settings for all inputs/variables.

Interpolation This comes from a lack of available data collected from computer model simulations and/or experimental measurements. For other input settings that don't have simulation data or experimental measurements, one must interpolate or extrapolate in order to predict the corresponding responses.